

Python Course for Physicists (And everyone else!)

Chunk 7: Imports

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Live Coding: Being Lazy - Let Others Code

- Imports allow you to use code, that others have programmed.
- „Package“ of code: module

- To use a module:

```
import math
print(math.sqrt(2))
```

1.4142135623730951

- Direct use of functions:

```
from math import sqrt
print(sqrt(2))
```

1.4142135623730951

Live Coding: More options:

- Import multiple functions:

```
from math import sqrt, sin, tan, cos
print(sqrt(2))
print(sin(2))
print(tan(2))
print(cos(2))
```

```
1.4142135623730951
0.9092974268256817
-2.185039863261519
-0.4161468365471424
```

- Import all:

→ Not recommended

```
from math import *
print(atanh(0.5))
```

```
0.5493061443340549
```

- Rename on import

```
from math import log as ln
from math import e
print(ln(e))
```

```
1.0
```

```
import math as m
print(m.log(m.e))
```

```
1.0
```

Built-in Modules → External Modules

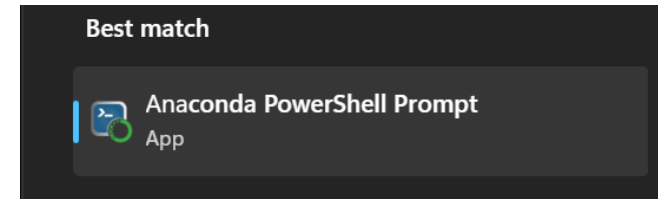
- `Math` is a built-in module in python
- Users upload their modules for use by other people
- For example: `numpy` $\triangleq$ numerical python
- External modules have to be installed in the command line
→ Otherwise:

```
import numpy
```

```
-----  
ModuleNotFoundError                                Traceback  
Cell In[16], line 1  
----> 1 import numpy  
  
ModuleNotFoundError: No module named 'numpy'
```

Tutorial: Installing Modules using Conda

- Open “Anaconda Powershell Prompt”



- Enter “conda install numpy”

In your case it will
show “base” instead
of “example”

```
(example) PS C:\Users\jona: conda install numpy
Channels:
  - defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\jona\miniconda3\envs\example

  added / updated specs:
    - numpy

The following packages will be downloaded:
```

- Type „y“ + Enter to proceed

```
vc14_runtime      pkgs/main/win-64::vc14_runtime-14.44.35208-h492
vs2015_runtime    pkgs/main/win-64::vs2015_runtime-14.44.35208-ha
wheel             pkgs/main/win-64::wheel-0.45.1-py313haa95532_0
xz                pkgs/main/win-64::xz-5.6.4-h4754444_1
zlib              pkgs/main/win-64::zlib-1.3.1-h02ab6af_0

Proceed ([y]/n)? |
```

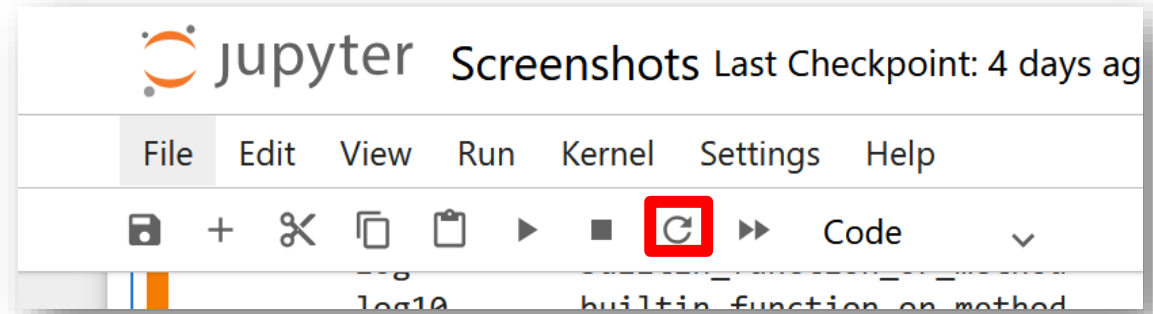
Optional: Check installation with “conda list”

- “conda list” lists all installed packages
- When installing numpy a lot of other packages get installed as well (dependencies)
- Note: “conda” and “pip” do very similar things → mostly interchangeable (Relevant when googling)

```
(example) PS C:\Users\jona> conda list
# packages in environment at C:\Users\jona\miniconda3\envs\example:
#
# Name                    Version            Build    Channel
blas                      1.0                mkl
bzip2                     1.0.8             h2bbff1b_6
ca-certificates           2025.9.9          haa95532_0
expat                     2.7.1             h8ddb27b_0
intel-openmp              2025.0.0          haa95532_1164
libffi                    3.4.4             hd77b12b_1
libmpdec                  4.0.0             h827c3e9_0
libzlib                   1.3.1             h02ab6af_0
mkl                       2025.0.0          h5da7b33_930
mkl-service               2.5.2             py313h0b37514_0
mkl_fft                  1.3.11            py313h5810407_1
mkl_random                1.2.8             py313hedd7022_1
numpy                     2.3.3             py313h050da96_0
numpy-base                2.3.3             py313h1e017a8_0
openssl                   3.0.18            h543e019_0
pip                       25.2              pyhc872135_0
python                    3.13.7            h260b955_100_cp313
python_abi                3.13              1_cp313
setuptools                72.1.0            py313haa95532_0
```

Using installed modules

- Restart kernel:



- Import works now:

```
import numpy
```

Good Luck with your Tasks 😊

ME:

```
import tensorflow as plt
import pandas as np
import numpy as tf
import matplotlib.pyplot as pd
```

Satan:



[https://www.reddit.com/r/ProgrammerHumor/comments/o3zjp4/thats_the_evilest_thing_i_can_imagine/]